LAB on Autonomous Driving Cars (700.385, 26S) (700.386, 256)

Final Project Guidelines

For every student, please follow the following instructions regarding the final project.

Instructions:

Use Carla simulator to implement the following:

1. **Lane Detection for Autonomous Driving Cars in the simulator:**
   a. Choose the map you want to work on using the (*init.py*) file.
   b. Using (*carla_lane_detection_student_version*) file, Implement the rendering part and generate *pygame* windows containing:
      i. Raw image
      ii. Semantic Segmentation image
      iii. Edge detection Image
      iv. Segmented lanes image
      v. Improved segmented lanes image (using Dilation or any image processing technique)

2. **Target point on segmented lanes image saved from the simulation run:**
   a. In colab notebook, implement the target point on one image saved from lane detection simulation; use the segmented lane image (masked image).
   b. File of (*Target_point_on_image.ipynb*) is uploaded to the final project folder

3. **Implement the target point in the CARLA simulator:**
   a. Use the (*Target_point_CARLA.py*) file to implement the *(lanes_detection*) function. And note that all functions used in the (*lanes_detection*) function are already implemented.
   b. Generate the pygame window for the target point in CARLA.

You need to deliver the following:

1. *.py* file for lane detection in CARLA	(task 1)	(40%)
2. *.ipynb* for target point on an image	(task 2)	(20%)
3. *.py* file for target point in CARLA	(task 3)	(40%)

All implemented codes should be explained in comment in the code file.

The final project is worth **50%** of the course weight. Submission is on Moodle.

Deadline: 30.07.2026
All grades will be submitted after the submission deadline

❖ For any questions related to the final project, please send me an email.
Email: Mohamed.salem@aau.at